

Statistical Modelling & Analysis of Rainfall Data Using Gamma and $M(\sigma, \gamma)$ Distributions

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Introduction

At a certain location, daily rainfall totals were measured (in millimetres, mm) for 40 years. Over this 40 year period, the daily total exceeded 25mm on 145 days. This investigation involves analysing a data frame, 'Rainfall', within R which contains these 145 excess amounts.

Gamma Model

I will model the data of the excess rainfall amounts as realisations of independent and identically distributed (iid) Gamma random variables, $Y_i \sim \text{Gamma}(\alpha, \beta)$, where α is the shape parameter and β is the rate parameter. The sample provided shows that the estimate of the mean is $\bar{y} \approx 18.9$, and the estimate of the variance is $s^2 \approx 616.1$. The estimates of the model parameters are found to be

$$\hat{\alpha} = \frac{\bar{y}^2}{s^2} = \frac{18.9^2}{616.1} = 0.579, \hat{\beta} = \frac{\bar{y}}{s^2} = \frac{18.9}{616.1} = 0.0307$$

The estimated standard error of $\hat{\alpha}$ is 0.107, and the estimated standard error of $\hat{\beta}$ is 0.00680.

As shown in Figure 1, the probability density function for the $\text{Gamma}(0.579, 0.0307)$ distribution is superimposed on the data distribution, and the model appears to approximate the data distribution relatively accurately. Although the model highly overestimates the occurrence of 0-5mm of excess rainfall, suggesting a lack of accuracy of the model when predicting smaller excess rainfall data, beyond 60mm of excess rainfall the model seems to fit well with the data. This is seen, as these rainfall values

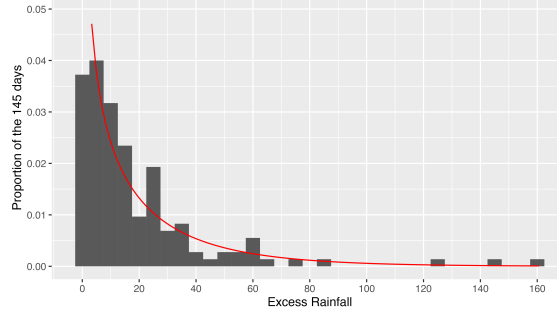


Figure 1: Shows the histogram of the excess rainfall data plotted with the pdf of the $Gamma(0.579, 0.0307)$ distribution (red line)

above 60mm occur on very few days, captured by the pdf of the model as it slowly approaches a proportion of 0. However, the model underestimates the likelihood of higher excess rainfall values around 123mm, 143mm and 160mm, indicating a lack of accuracy for higher excess rainfall values. Overall, the shape of the pdf roughly fits with the shape of the histogram, suggesting the model's realism is fairly accurate.

$M(\sigma, \gamma)$ Model

Now I will model the excess rainfall amounts as realisations of iid random variables with a distribution denoted by $M(\sigma, \gamma)$. In this model there are two parameters, σ and γ , where $\sigma > 0$, and $\gamma < 1/2$. From the equations below we obtain the values of the model parameter estimates to be $\hat{\sigma} = 14.9$ and $\hat{\gamma} = 0.210$;

$$18.9 = \frac{\hat{\sigma}}{1 - \hat{\gamma}}, 616.1 = \frac{\hat{\sigma}^2}{(1 - \hat{\gamma})^2(1 - 2\hat{\gamma})}$$

The estimated standard error of $\hat{\sigma}$ is 1.872, and the estimated standard error of $\hat{\gamma}$ is 0.0892.

This new model seems to be a good approximation of the data distribution. The fitted pdf of the model in Figure 2 closely matches up with the actual rainfall data from around 8mm - 18mm of excess rainfall, but, similar to the model in Figure 1 this model underestimates excess rainfall data between 22.5mm and 27.5mm, as well as having the same accuracy as the model in Figure 1 for values above 60mm.

Comparing the two models

Both models have two parameters, indicating that their simplicity is the same. For both, it could be argued that if the data was taken over a longer period of time, then the models could be a realistic fit for the data as the shape of the model's pdfs

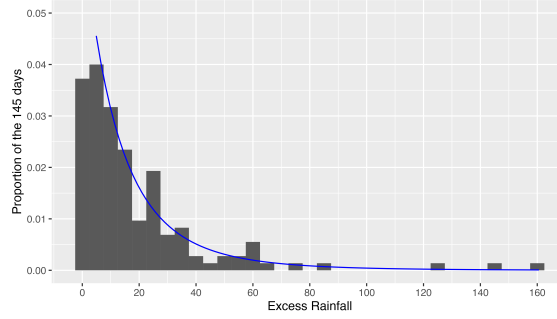


Figure 2: Shows the histogram of the excess rainfall data plotted with the pdf of the new $M(14.9, 0.210)$ distribution (blue line)

are similar to the general shape of the histogram. Overall, I prefer the first model. Although the pdf of the second model matches up slightly closer to data points on the histogram, the estimates of the standard errors of the parameter estimates are lower in the first model. The smaller standard errors in the first model indicate that the model fits the data better, and generally smaller standard errors provide more precise estimates of parameters.

Using the preferred model of $Gamma(0.579, 0.0307)$, the 10-year return level is calculated to be 110.98, meaning the daily rainfall total of 110.98mm is exceeded once every 10 years on average, according to this model.

Statistical Analysis of a Study Assessing the Effectiveness of a New Antibiotic

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Introduction

A study was carried out over one year, and aimed to determine the effectiveness of a new antibiotic for treating a certain type of infection in hospital patients in the UK. Patients with the infection in two hospitals in England had to consent to be involved in the study. This involved patients taking the new antibiotic, and following this doctors determined whether the infection cleared within two weeks. The data is modelled as iid Bernoulli random variables, $X_i \sim \text{Bernoulli}(p)$, where p is the probability parameter, i.e. p represents the proportion of patients who are cured within two weeks.

Assessing the effectiveness of the new antibiotic

The point estimate of p is $\hat{p} = 0.735$, i.e. the point estimate for the effectiveness of the new antibiotic is 73.5%. As this distribution is non-Normal with an unknown variance, s^2 , and a large sample size, n , the 95% confidence interval for this point estimate $\hat{p}$ is given by $\bar{x} \pm z_p s / \sqrt{n}$, where z_p is the 0.975-quantile of the standard Normal distribution, $N(0, 1)$. So, the 95% confidence interval for the probability that a patient from either hospital is cured within two weeks after using the new antibiotic is $0.735 \pm 1.96(\sqrt{0.197})/\sqrt{98} = (0.647, 0.823)$. This confidence interval suggests that 95% of such intervals contain the true value of p .

The new antibiotic may not be much more effective compared to the existing antibiotic, as the confidence interval $(0.647, 0.823)$ includes 70%, the effectiveness rate of the existing antibiotic. As a result, we cannot draw a certain conclusion that the new antibiotic is more effective.

The confidence intervals were calculated for each of the 100 samples. The results are shown in Figure 3.

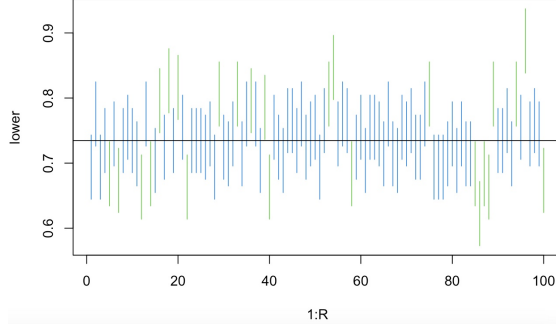


Figure 3: 73.5% confidence intervals (vertical lines) for p computed for each of the 100 samples. Some intervals contain the true value of the estimator (blue) and some do not (green). The horizontal line represents the point estimate of p

Assessing the effectiveness of the new antibiotic in Hospital A and B

Now, we compare the effectiveness of the new antibiotic in both hospitals, where Hospital A is a general hospital, but Hospital B is a specialist hospital that treats only a certain type of patient.

We can construct a confidence interval for both hospitals separately, in the same way that we used above. The point estimate of p for Hospital A is $\hat{p} = 0.836$, and for Hospital B it is $\hat{p} = 0.605$. The 95% confidence interval for Hospital A for its point estimate is given by $\bar{x} \pm z_p s / \sqrt{n} = 0.836 \pm 1.96(\sqrt{0.139}) / \sqrt{98} = (0.762, 0.910)$. The 95% confidence interval for Hospital B for its point estimate is given by $\bar{x} \pm z_p s / \sqrt{n} = 0.605 \pm 1.96(\sqrt{0.245}) / \sqrt{98} = (0.507, 0.703)$.

The 95% confidence interval for Hospital A's point estimate suggests that the new antibiotic is more effective than the existing antibiotic for patients that are treated at Hospital A, as the confidence interval lies entirely above the 70% known effectiveness of the previous antibiotic. However, the 95% confidence interval for Hospital B's point estimate suggests that the new antibiotic is less effective than the existing one for patients treated at Hospital B, as the confidence interval lies almost entirely below the 70% known effectiveness of the previous antibiotic. Overall, the effectiveness of the new antibiotic does seem to differ between the two hospitals, with it being more effective for patients in Hospital A. As Hospital A is a general hospital, this suggests that the new antibiotic works better on the general population. However, the antibiotic seems to be less effective with patients in Hospital B, suggesting that it does not work as well as the existing antibiotic in treating patients of a certain type.

Strengths of the design of the study

One strength of the study design is its duration. The study lasted for one year, allowing for a sufficient period of time for the researchers to test and analyse results on the effectiveness of the new antibiotic. Another strength is that the study involved a comparison between the effectiveness of the new antibiotic on different types of patients, allowing us to see how its effectiveness changes amongst varying patient populations. Also, the study was carried out on consenting patients only, showing a willingness to participate was considered by the researchers.

Weaknesses of the design of the study

One weakness to the study is that it only uses patients from two hospitals in England. This may mean that there is some generalisation bias, making the study population unrepresentative of the target population. Also, there may be confounding factors that have not been considered in the study. For example, Hospital B treats only a certain type of patient, and these patients may all have something in common that constitutes them as a "certain type of patient," such as an underlying health condition. This may affect their response to the new antibiotic, meaning any inferences made about these patients may be inaccurate. Furthermore, there is a sampling bias, as the study involves only consenting patients; a volunteer bias may arise as a result if the volunteers differ systematically from others in the population.

Overall conclusion of the effectiveness of the antibiotic

The results suggest that the new antibiotic is estimated to be more effective in clearing infections within two weeks for patients in Hospital A compared with Hospital B. However, the confidence intervals calculated above suggest uncertainty in the parameter estimates of p , as they are quite wide. This could be due to the weaknesses of the study that have been discussed. Therefore, although the study suggests some improvements in its effectiveness, a study involving more patients from more hospitals, with a greater variety of health conditions is necessary in order to be confident in this inference.